

R for bioinformatics, data visualisation, part 1

HUST Bioinformatics course series

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Section 1

section 1: TOC

前情提要

iterations 与 并行计算

- for loop
- apply functions
- dplyr 的本质是 遍历
- map functions in purrr package
- 遍历 与 并行计算

相关包

- purrr
- parallel
- foreach
- iterators

本次提要

- basic plot functions
- basic ggplot2
- special letters
- equations
- advanced ggplot2

Section 2

section 2: basic plot functions using R

R basic plot functions

基础做图由 plot 提供。

使用系统自带的 `swiss` 数据举例，它包含了 47 个法语地区的一些社会经济指标。

```
head(swiss);
```

	Fertility	Agriculture	Examination	Education	Catholic
Courtelary	80.2	17.0	15	12	9.96
Delemont	83.1	45.1	6	9	84.84
Franches-Mnt	92.5	39.7	5	5	93.40
Moutier	85.8	36.5	12	7	33.77
Neuveville	76.9	43.5	17	15	5.16
Porrentruy	76.1	35.3	9	7	90.57

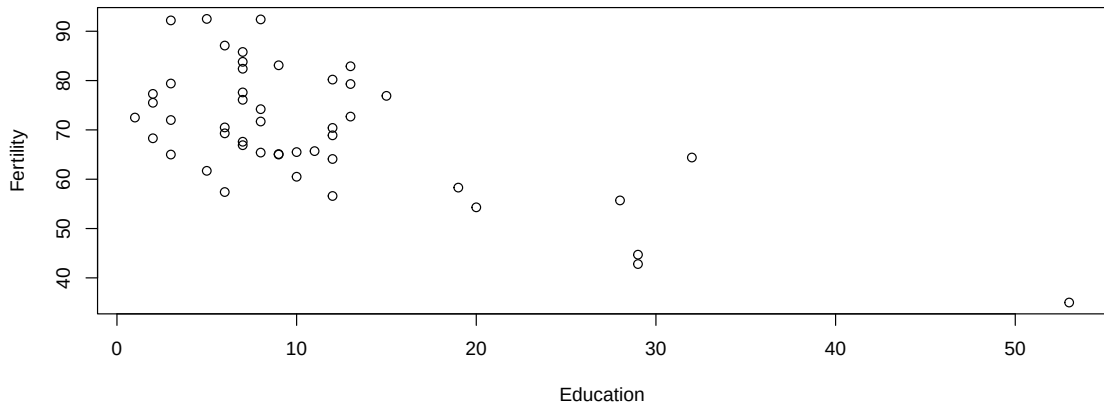
	Infant.Mortality
Courtelary	22.2
Delemont	22.2
Franches-Mnt	20.2
Moutier	20.3
Neuveville	20.6
Porrentruy	26.6



散点图 (dot plot)

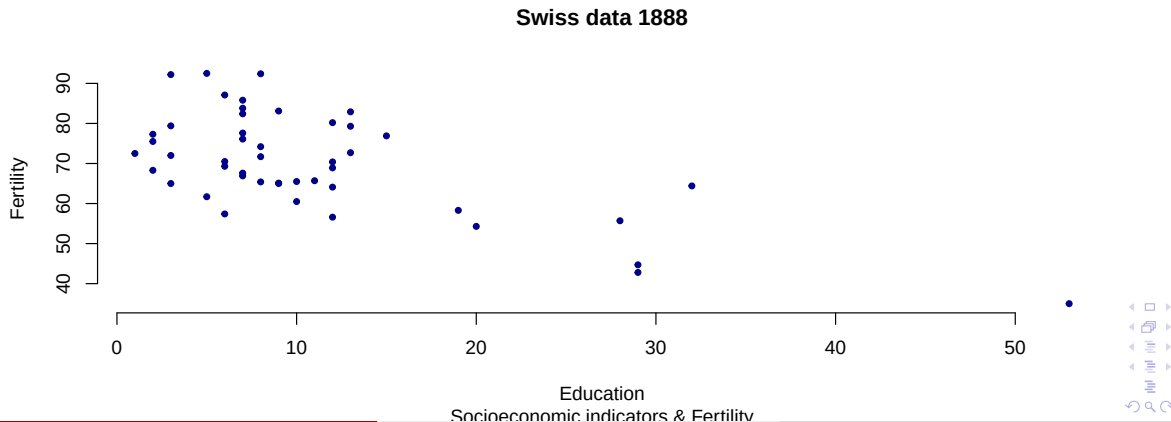
我们看一下 教育 与生育率的关系:

```
with( swiss, plot( Education, Fertility ) );
```



plot 的参数初探：先看示例

```
with( swiss, plot(Education, Fertility, type = "p", main = "Swiss data 1888",
  sub = "Socioeconomic indicators & Fertility",
  xlab = "Education", ylab = "Fertility", col = "darkblue",
  xlim = range( Education ), ylim = range( Fertility ),
  pch = 20, frame.plot = F) );
```



plot 参数, an annotated example

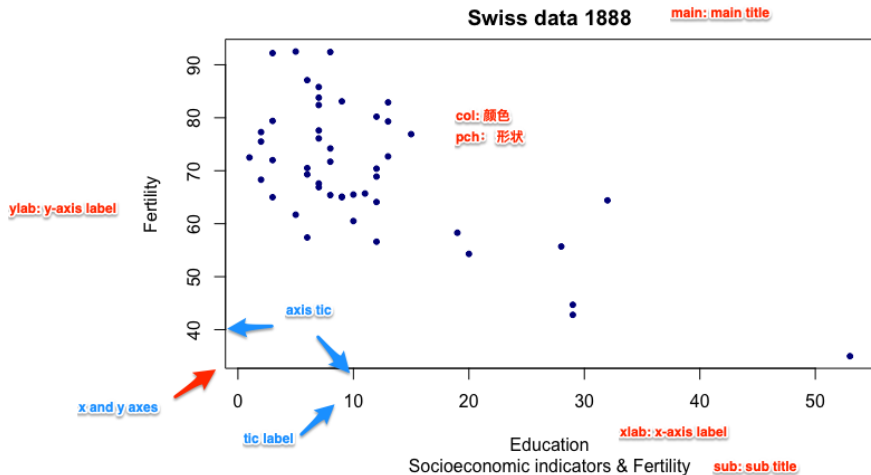
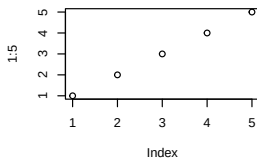


Figure 1: an annotated example

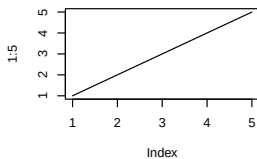
plot 支持的画图类型, 参数 `p = '?'` 的取值

```
par( mfrow = c(2,4) ); ### 在一张图上画 2 x 4 个 panel
opts <- c( "p", "l", "o", "b", "c", "s", "S", "h" );
for( o in opts ){
  plot(1:5, type = o, main = paste( "type=", o ) );
}
```

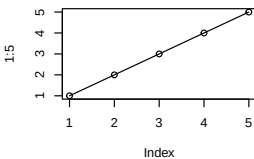
type= p



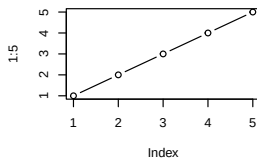
type= l



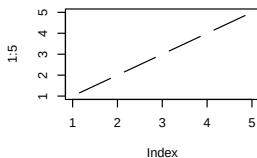
type= o



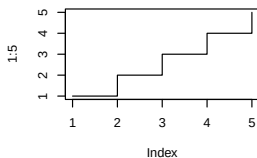
type= b



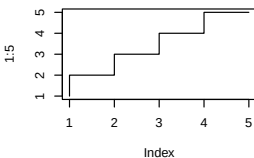
type= c



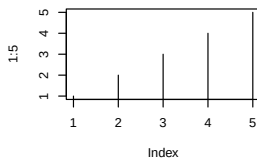
type= s



type= S



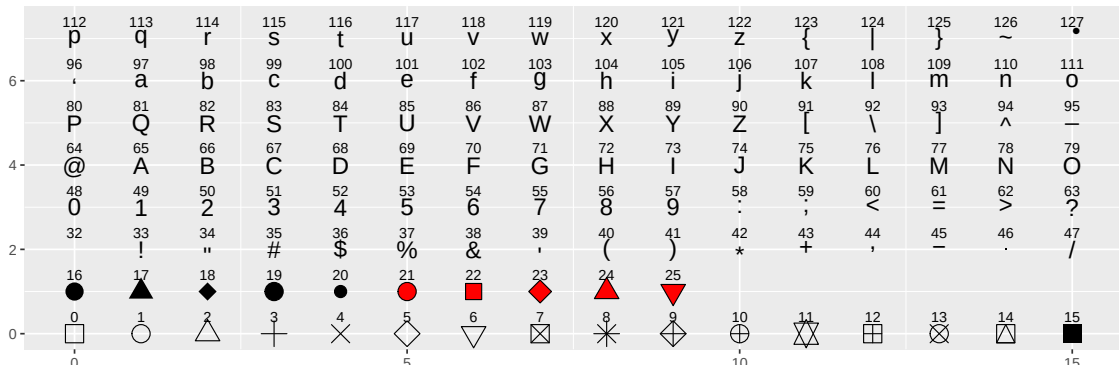
type= h



pch 是什么?

决定了数据点的形状, 注意它的取值范围

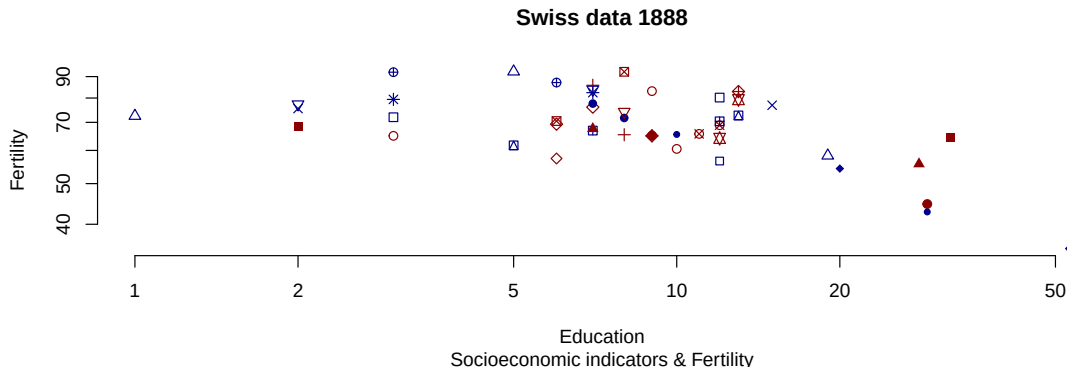
```
library(tidyverse); library(extrafont);
ggplot( data.frame( p = c(0:25, 32:127) ) ) +
  scale_y_continuous( name = "" ) + scale_x_continuous( name = "" ) +
  scale_shape_identity() +
  geom_point( aes( x = p%%16, y = p%%16, shape = p ), size = 5, fill = "red" ) +
  geom_text( aes( x = p %% 16, y = p%%16 + 0.4, label = p ), size = 3 );
```



log transform axes

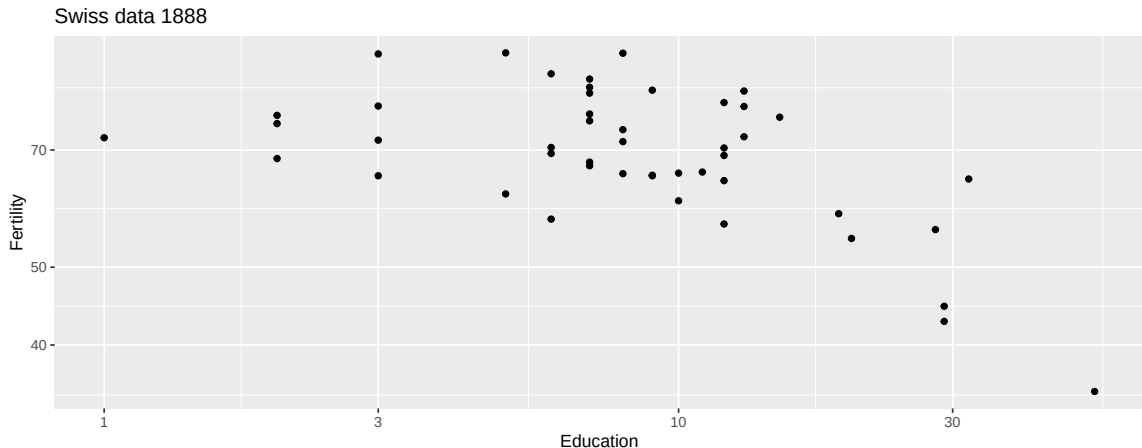
plot 还有一些其它有用的参数, 详见: ? plot.default

```
with( swiss, plot(Education, Fertility, type = "p", main = "Swiss data 1888",
  sub = "Socioeconomic indicators & Fertility",
  xlab = "Education", ylab = "Fertility", col = c("darkblue", "darkred"),
  xlim = range( Education ), ylim = range( Fertility ),
  pch = 0:25, frame.plot = F, log = "xy") );
```



ggplot 版本

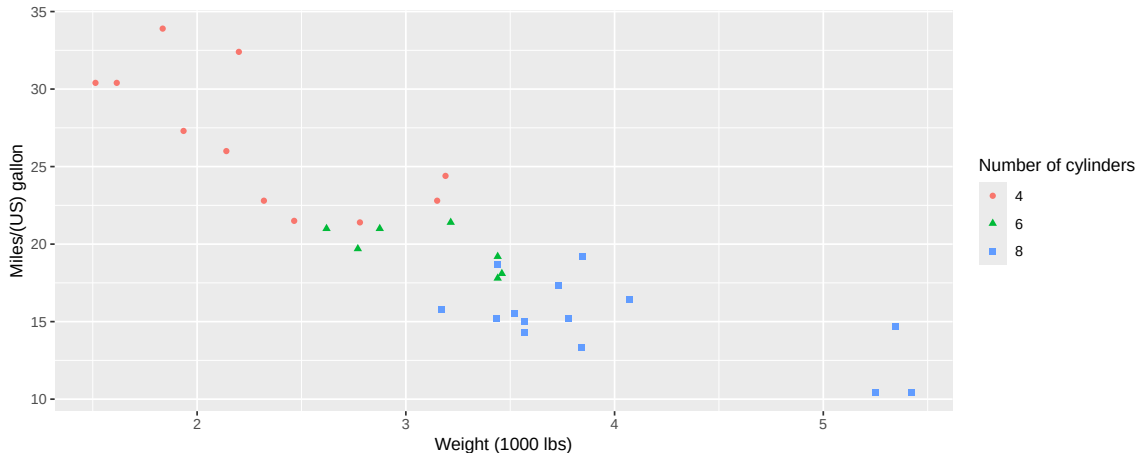
```
ggplot( swiss, aes( x = Education, y = Fertility ) ) +  
  geom_point( ) + scale_x_log10() + scale_y_log10() +  
  xlab( "Education" ) + ylab( "Fertility" ) + ggtitle( "Swiss data 1888" );
```



ggplot 更多散点示例

以 mtcars 为例

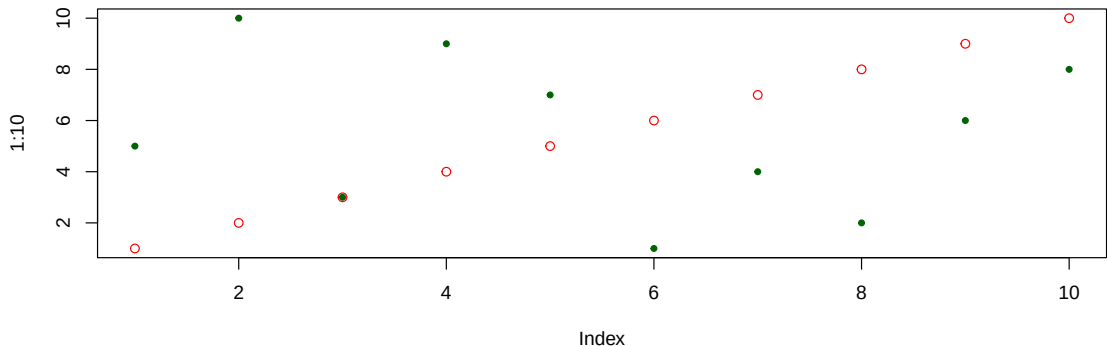
```
ggplot( mtcars, aes( x = wt, y = mpg, colour = factor( cyl ), shape = factor(cyl) ) ) +
  geom_point() + xlab( "Weight (1000 lbs)" ) + ylab( "Miles/(US) gallon" ) +
  labs( colour = "Number of cylinders", shape = "Number of cylinders" );
```



plot: high-level vs. low-level plots

- **high level:** plotting functions create a new plot on the graphics device
- **low level:** plotting functions add more information to an existing plot

```
plot( 1:10, col = "red" ); ## high level  
points( sample(1:10, 10), col = "darkgreen", pch = 20 ); ## low level
```



low level plots 列表

- points : 点图
- lines : 线图
- abline : 直线
- polygon : 多边形
- legend : 图例
- title : 标题
- axis : 轴 ...

high level plots 列表

- plot : 通用画图函数
- pairs
- coplot
- qqnorm
- hist
- dotchart
- image
- contour ...

注: 可以用 `add = TRUE` 参数 (如果可用) 将 high level 函数强制转换为 low level

图形相关参数（系统函数）

`par()` 函数：显示或修改当前图形设备的参数。用以下命令查看支持的内容：

```
par( c( "mar", "bg" ) ); ## 显示指定参数的值
```

```
$mar
[1] 5.1 4.1 4.1 2.1
```

```
$bg
[1] "transparent"
```

```
## 显示所有参数
par();
```

```
$xlog
[1] FALSE
```

```
$ylog
[1] FALSE
```

```
$adj
[1] 0.5
```

调整 `par()` 参数前请备份

`par()` 用于指定全局参数，因此在改变前尽量备份

```
oldpar <- par(); ## 备份  
  
## do some changes here ...  
  
## 恢复  
par( oldpar );
```

常用图形参数及调整: `margin`

图形边距 (figure margins)

```
par( mar = c( 5.1, 4.1, 4.1, 2.1 )); ## 设置新 margin
```

分别指定 下 -> 左 -> 上 -> 右 的边距, 即从下面开始, 顺时针移动。

单位是: text lines

或:

```
par( mai = c( 5.1, 4.1, 4.1, 2.1 )); ## 设置新 margin
```

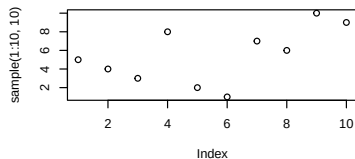
单位是: inch

常用图形参数及调整：多 panel

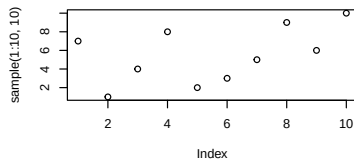
画 2x3 共 6 个 panel，从左到右。（2 行 3 列）

```
par( mfrow=c(2,3) );
for( i in 1:6 )
  plot( sample( 1:10, 10 ), main = i );
```

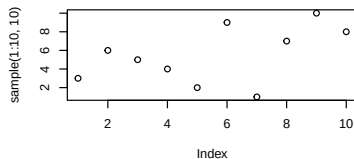
1



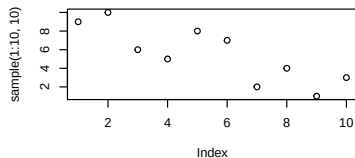
2



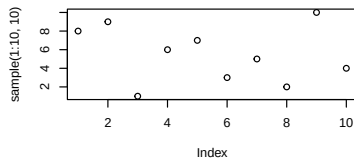
3



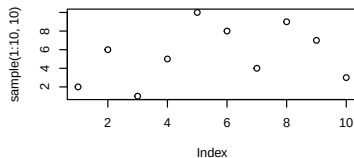
4



5



6

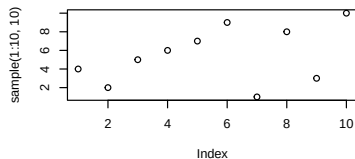


常用图形参数及调整: 多 panel , cont.

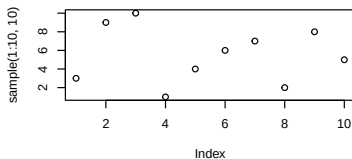
画 2x3 共 6 个 panel, 从上到下。(2 行 3 列)

```
par( mfcol=c(2,3) );
for( i in 1:6 )
  plot( sample( 1:10, 10 ), main = i );
```

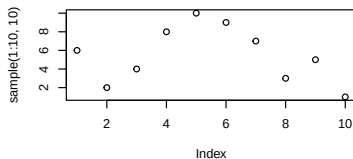
1



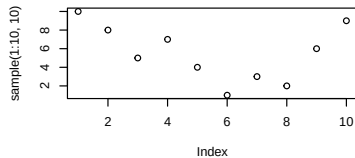
3



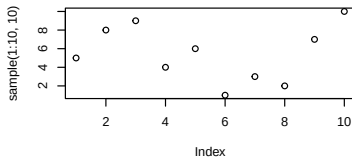
5



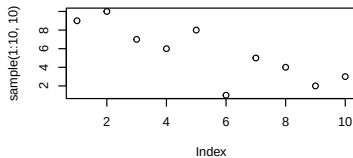
2



4



6



重要概念：图形设备

图形设备是指图形输出的设备，可以将图形设备理解为保存格式。

默认设备是：

- `X11()` : *nix
- `windows()` : windows
- `quartz()` : OS X

图形显示在显示器上。

图形设备: cont.

常用其它设备有:

- pdf()
- png()
- jpeg()

分别对应输出文件格式。

常用图形设备: pdf()

使用方法如下:

```
## 创建一个新设备/ pdf 文件
pdf( file = "/path/to/dir/<file_name>.pdf", height = 5, width = 5 );
plot(1:10); ## 作图;
dev.off(); ## 关闭设备
```

说明

- ① 默认文件名为 Rplots.pdf ,
- ② dev.off() 必须关闭。关闭后, 返回到最近使用的图形设备
- ③ height 和 width 参数的单位是 inch
- ④ 如果运行多个 high level 作图命令, 则会产生多页 pdf

请尽量使用 pdf 作为文件输出格式

- ① 生信图片大多是点线图，适合保存为矢量格式（如 pdf, ps 等）；
- ② 矢量图可无限放大而不失真（变成像素）；
- ③ 可由 Adobe Illustrator 等矢量图软件进行编辑

Section 3

section 3: ggplot2 基础

一个简单的示例

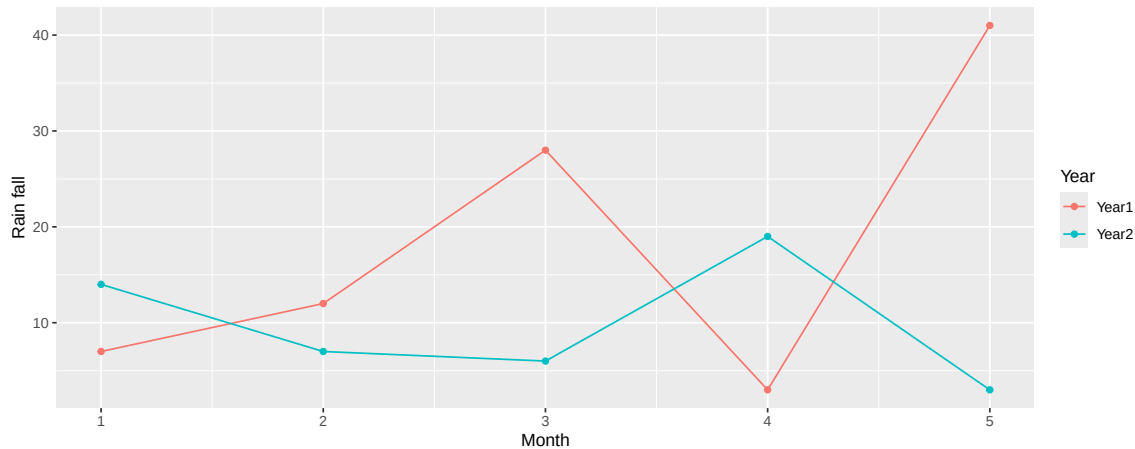
```
## 准备数据
year1 <- c(7, 12, 28, 3, 41);
year2 <- c(14, 7, 6, 19, 3);

df <- rbind(
  tibble( month = 1:length( year1 ), value = year1, cat = "Year1" ),
  tibble( month = 1:length( year2 ), value = year2, cat = "Year2 " ) );

plot1 <- ## 将图保存在变量中;
  ggplot( df, aes( x = month, y = value, colour = cat ) ) +
  geom_line() + geom_point() +
  xlab( "Month" ) + ylab( "Rain fall" ) +
  labs( colour = "Year");
```

画图

```
plot1; ## 画图
```



ggplot2 基础概念详解

```

ggplot( df, aes( x = month, y = value, colour = cat) ) +
  geom_line() + geom_point() +
  xlab( "Month" ) + ylab( "Rain fall" ) +
  labs( colour = "Year" );

```

Figure 2: ggplot2 参数简介

- ① **aes** (aesthetics) 美学: 控制全局参数, 包括: x,y 轴使用的数据, 颜色 (colour, fill) , 形状 (shape) , 大小 (size) , 分组 (group) 等等;
- ② 图层: `geom_<layer_name>` ; 每张图可有多多个图层 (此处有两个); 图层可使用全局数据 (df) 和参数 (aes), 也可以使用自己的 aes 和 数据;
- ③ 其它控制函数

ggplot2 优缺点

ggplot2 优点:

- ① 强大又专业
- ② 复杂又好看
- ③ canvas 大小, 坐标会根据数据、图层自动调整, 让用户专注于作图本身;

缺点:

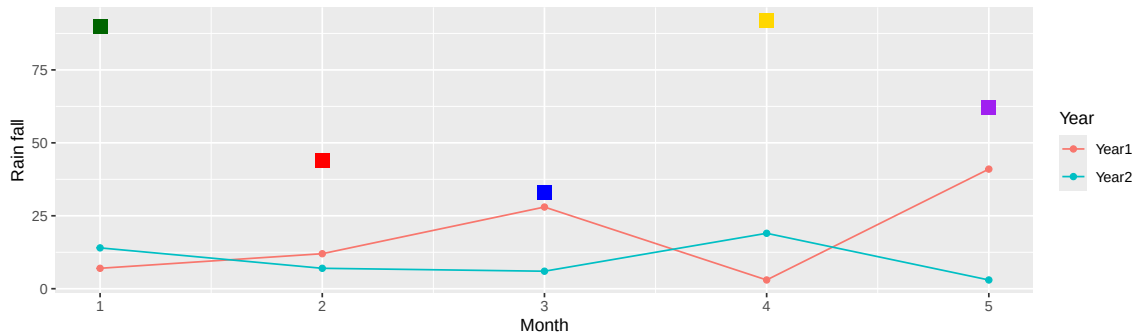
太难学 !

图层使用自己的数据， 示例

图层可从 ggplot 继承数据 和 其它参数， 包括 aes， 但也可使用自己的

plot1 +

```
geom_point( data = data.frame( x2 = 1:5, y2 = sample(30:100, 5) ),
  ## 注意: data = 是必须的
  aes( x = x2, y = y2 ), ## 使用自己的 aes ...
  colour = c("darkgreen", "red", "blue", "gold", "purple" ) ,
  shape = 15, size = 4 )
```



Section 4

section 4: ggplot2 作图的四个基本组成部分详解

1. 图层

`geom_< 图层名 >`

- `geom_point`, `geom_line`: 点线图, 用于揭示两组数据间的关系;
- `geom_smooth`: 常与 `geom_point` 联合使用, 揭示数据走势
- `geom_bar`: bar 图
- `geom_boxplot`: 箱线图, 用于比较 N 组数据, 揭示区别
- `geom_path`: 与 `geom_line` 相似, 但也可以画其它复杂图形
- `geom_histogram`, “`geom_density`”: 数据的分布, 也可用于多组间的比较
- 其它十余种, 请见 “ggplot2: elegant graphics for data analysis” 一书 !!

2. scale 显示控制

基本规则

- `scale_< 控制内容 >_< 控制手段 >`, e.g. `scale_color_manual()`: 以手选方式 控制 颜色

四种控制类型

- `scale_color_...`
- `scale_fill_...`
- `scale_shape_...`
- `scale_size_...`

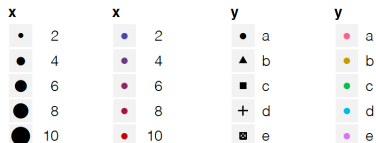
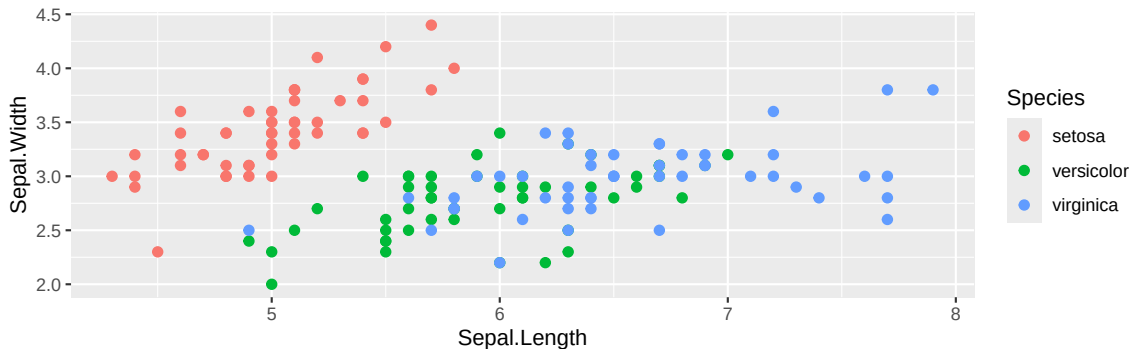


Figure 3: 数据的 4 种 scale 方法

2.1 改变颜色

```
library(ggplot2);  
ggplot(iris, aes(x=Sepal.Length, y=Sepal.Width, color=Species)) +  
  geom_point(size=2);
```



上图的解释

- 使用 `iris` 里 萼片 大小数据;
- `colour = Species`: 根据 `Species` 列决定颜色;
- 共需要 3 种 颜色;
- 由默认色板决定;

默认的色板

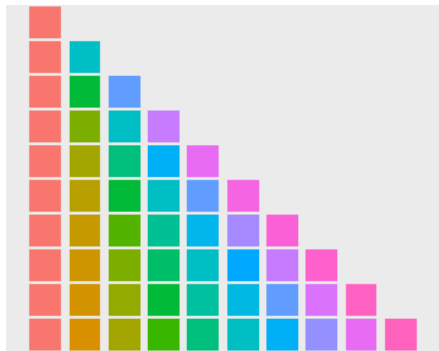
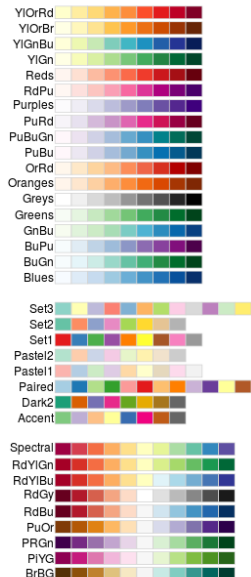


Figure 4: default discrete colour palette

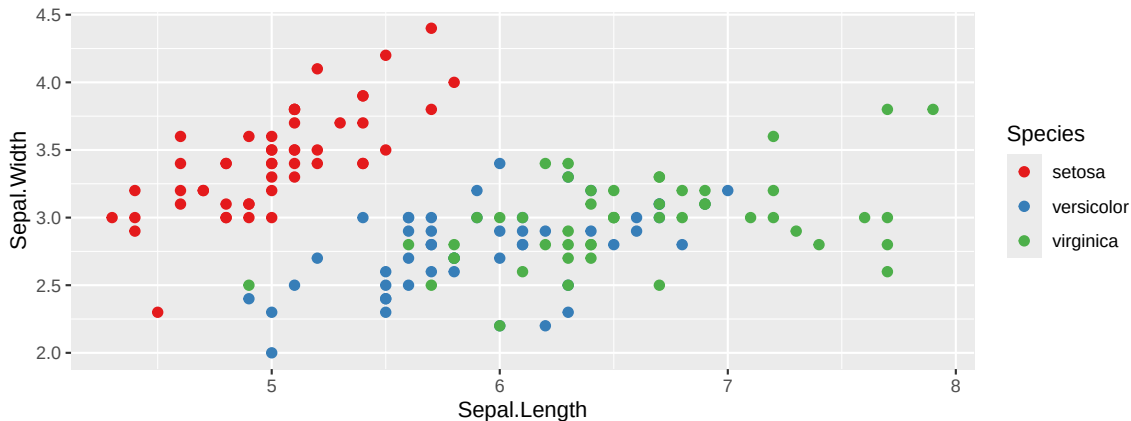
- 根据需要的颜色数量，取相应的行

使用其它色板，比如 `scale_colour_brewer()`



scale_colour_brewer(), 使用示例

```
ggplot(iris, aes(x=Sepal.Length, y=Sepal.Width, color=Species)) + geom_point(size=2) +  
  scale_color_brewer( palette = "Set1" );
```



- 使用方法: `scale_color_brewer( palette = "<palette name>" );`
- 取前 N 个颜色

scale_color_brewer 提供的其它色板

Diverging

BrBG, PiYG, PRGn, PuOr, RdBu, RdGy, RdYIBu, RdYlGn, Spectral

Qualitative

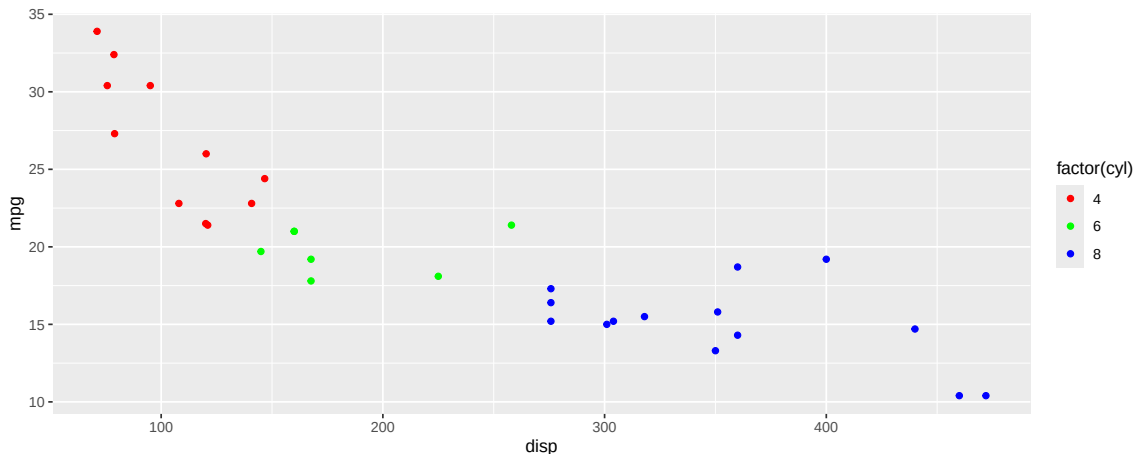
Accent, Dark2, Paired, Pastel1, Pastel2, **Set1**, Set2, Set3

Sequential

Blues, BuGn, BuPu, GnBu, Greens, Greys, Oranges, OrRd, PuBu, PuBuGn, PuRd, Purples, RdPu, Reds, YlGn, YlGnBu, YlOrBr, YlOrRd

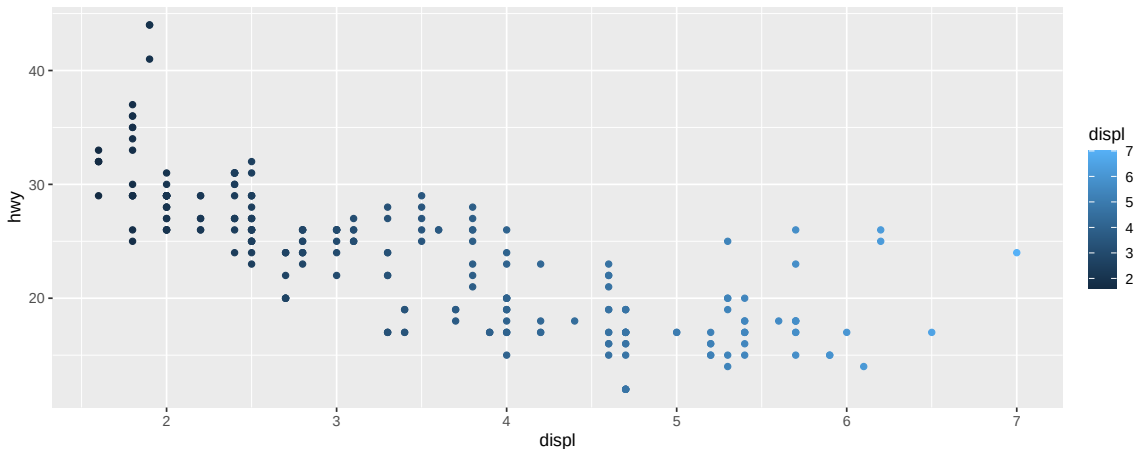
手动设置颜色: `scale_colour_manual()`

```
mtcars %>% ggplot( aes(displ, mpg) ) + geom_point( aes( color = factor(cyl) ) ) +  
  scale_color_manual( breaks = c("4","6","8"), values = c("red","green","blue") );
```



如何对应 数量变量 与颜色？使用 渐变颜色！

```
mpg %>% ggplot( aes(displ, hwy) ) + geom_point( aes( color = displ ) );
```



- 根据 `displ` 排气量 的值显色;
- 默认使用 2 色;

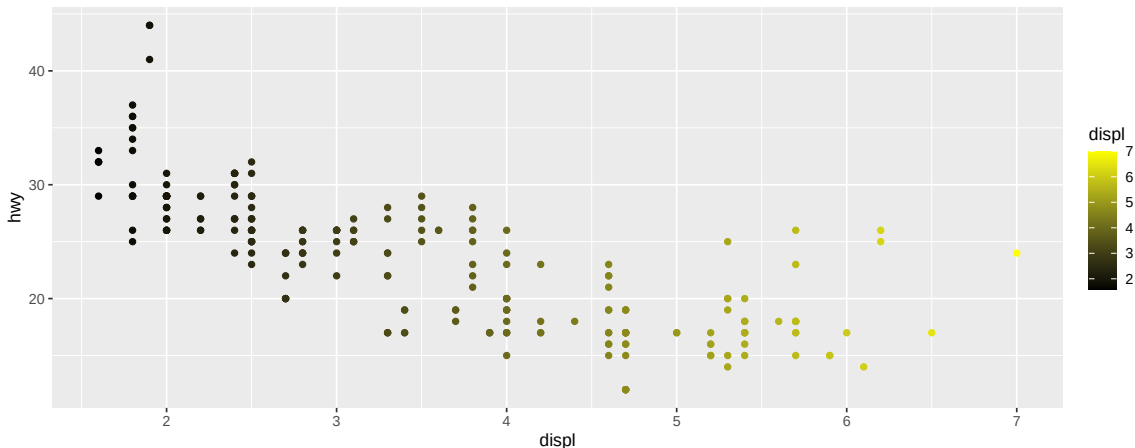
用 `scale_color_gradient()` 改变渐变颜色

```
scale_colour_gradient(  
  ...,  
  low = "#132B43",  
  high = "#56B1F7",  
  space = "Lab",  
  na.value = "grey50",  
  guide = "colourbar",  
  aesthetics = "colour"  
)
```

改变 low 和 high 的值即可;

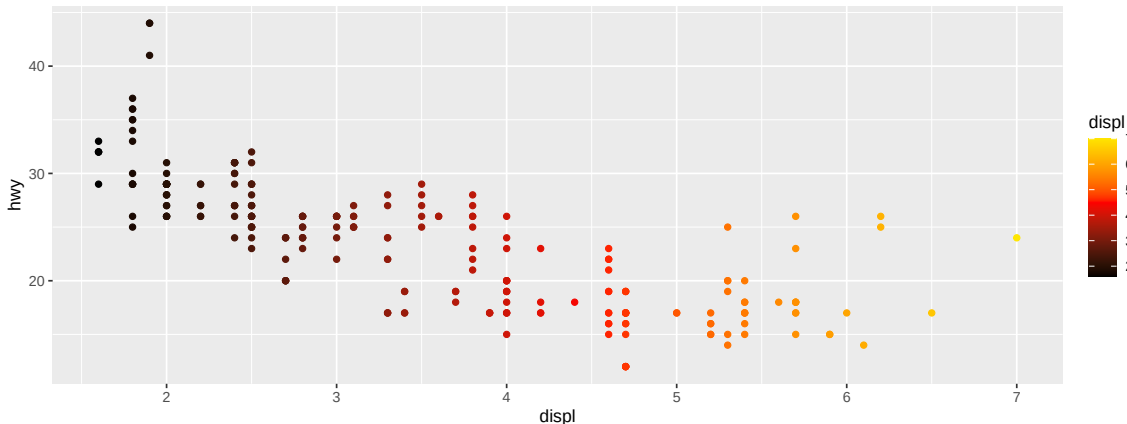
scale_color_gradient() 举例

```
mpg %>% ggplot( aes(displ, hwy) ) + geom_point( aes( color = displ ) ) +  
  scale_color_gradient( low = "black", high = "yellow" );
```



使用 3 个渐变色? 用 `scale_color_gradient2()`

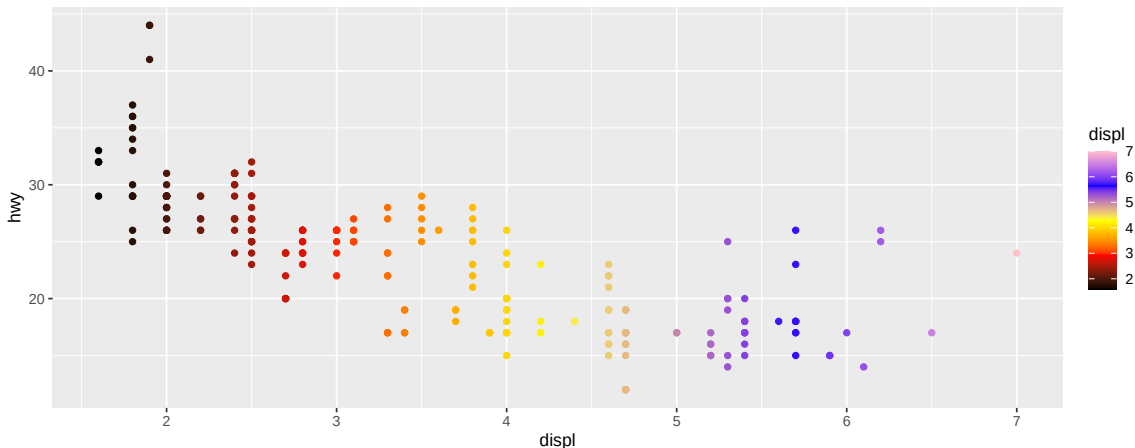
```
mpg %>% ggplot( aes(displ, hwy) ) + geom_point( aes( color = displ ) ) +  
  scale_colour_gradient2( low = "black", mid = "red", high = "yellow", midpoint = 4.5 );
```



- 三个颜色: low mid high
- 注意 midpoint 参数 !!

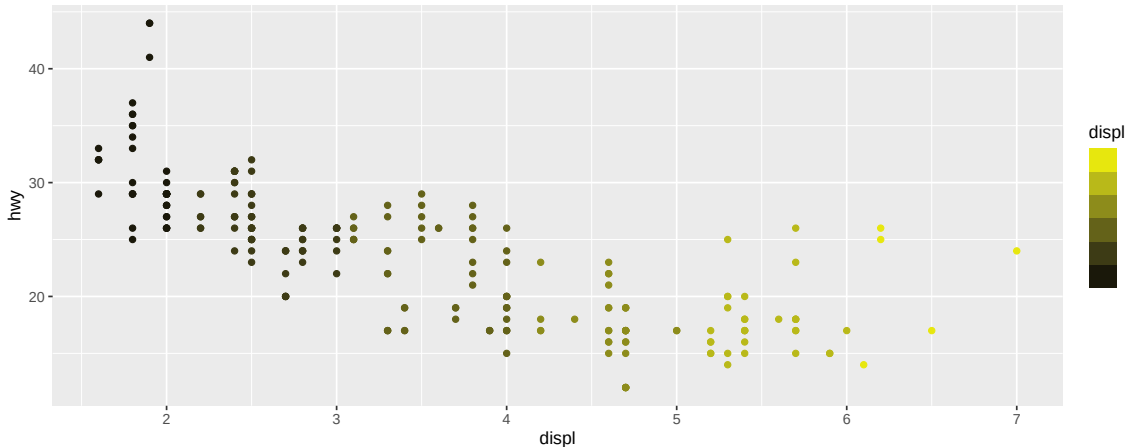
更多颜色？使用 `scale_colour_gradientn`

```
mpg %>% ggplot( aes(displ, hwy) ) + geom_point( aes( color = displ ) ) +
  scale_colour_gradientn( colors = c("black", "red", "yellow", "blue", "pink") );
```



一种有趣的渐变色显示方法: `scale_colour_binned`

```
mpg %>% ggplot( aes(displ, hwy) ) + geom_point( aes( color = displ ) ) +
  scale_color_binned( low = "black", high = "yellow" );
```



- 渐变色, 但分组显示

2.2 fill 颜色

fill 与 colour 有什么区别???

- colour defines the colour with which a geom is **outlined** (the shape's "stroke")
- fill defines the colour with which a geom is **filled**

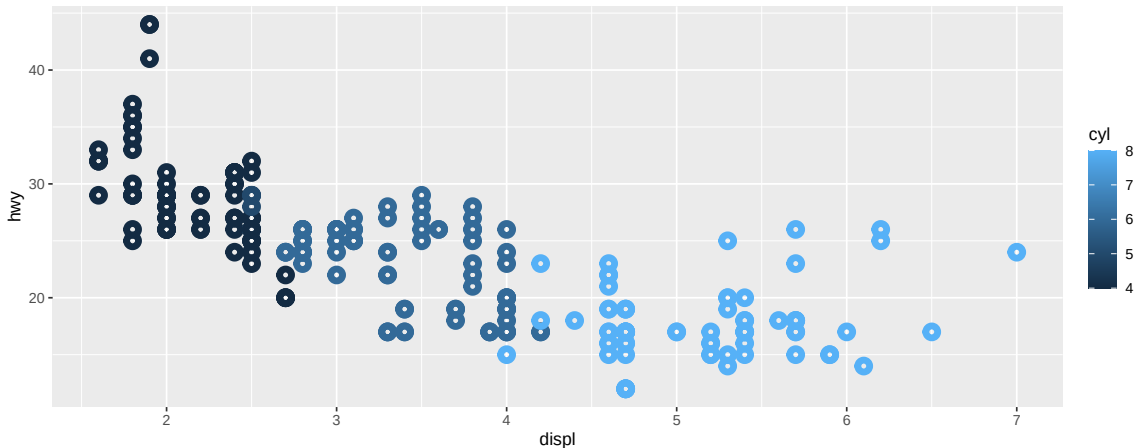
哪些 pch 们有 fill ??

- Points generally only have a colour and **no fill**
- However, point shapes **21–25** that include **both** a colour and a fill.

fill 与 colour 有啥区别??? (again)

找一个有两种属性的 point (pch = 21), 先看 fill 的效果;

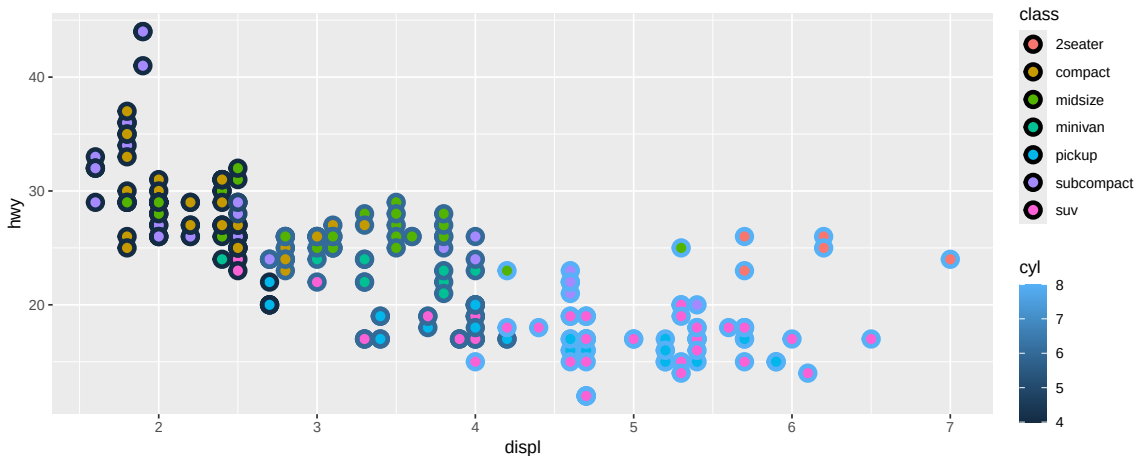
```
mpg %>% ggplot( aes(displ, hwy) ) +  
  geom_point( aes( colour = cyl ), shape = 21, stroke = 3);
```



fill 与 colour 的区别???, cont.

同时有 fill 和 colour

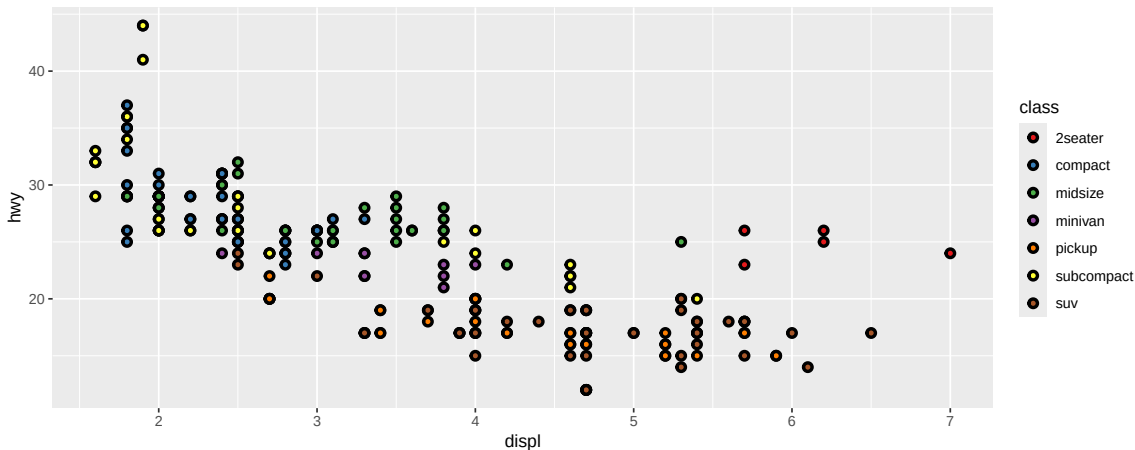
```
mpg %>% ggplot( aes(displ, hwy) ) +  
  geom_point( aes( colour = cyl, fill = class ),  
             shape = 21, size = 3, stroke = 2); ## 注意 shape 等参数在 aes() 内部与外部的区别
```



fill 的调色板 与 colour 一样！

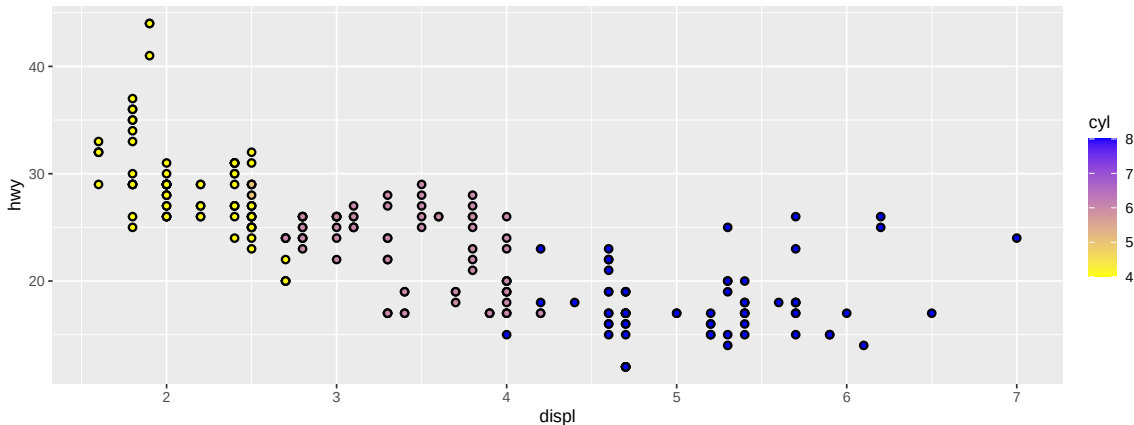
先看看离散值的色板

```
mpg %>% ggplot( aes(displ, hwy) ) +  
  geom_point( aes( fill = class ), shape = 21, stroke = 1.5) +  
  scale_fill_brewer( palette = "Set1" ) ##
```



fill 的 gradient 色板

```
mpg %>% ggplot( aes(displ, hwy) ) +  
  geom_point( aes( fill = cyl ), shape = 21, stroke = 1) +  
  scale_fill_gradient( low = "yellow", high = "blue" ) ;
```



注: 默认的 stroke 颜色为 black;

fill 与 colour 小结

① 做到两个匹配，即：

- a) aes 参数 (如 fill =) 与 函数 (如 scale_fill_xxx) 匹配; colour 也是如此;
- b) 数据类型与色板类型匹配: 数量配渐变, 离散配离散;

```
mpg %>% ggplot( aes(x=displ, y=hwy) ) +  
  geom_point( shape = 21, aes( fill = factor(cyl) ) ) +  
  scale_fill_brewer( palette = "Set2" );
```

- ② 两者色板是通用的; 函数也是, 比如 scale_fill_xx 对应 scale_colour_xx;
- ③ 有些 shape / pch 只有 colour (stroke), 有些则两者都有;

其它颜色相关函数

- `scale_fill_continuous`, `scale_colour_continuous`
- `scale_fill_manual`, `scale_colour_manual`

...

其它包中的调色板和相应函数

included in ggplot2

`scale_color_hue`, `scale_color_manual`, `scale_color_grey`, `scale_colour_viridis_d`,
`scale_color_brewer` ...

from the RColorBrewer package

`scale_color_brewer(palette = "<palette name>")` ... note: 函数属于 ggplot2

from the viridis package

`scale_color_viridis( discrete=TRUE, option="<palette name>" )` note: 提供了函数和 palette

other packages ...

- paletteer package: `scale_color_paletteer_xx` functions
- ggsci package

ggsci: 论文发表用的色板!!!

install

```
install.packages("ggsci"); # Install ggsci from CRAN:  
devtools::install_github("nanxstats/ggsci"); # or from github
```

contents

scale_color_<journal> 和 scale_fill_<journal> functions and color palettes

supported journals

- NPG scale_color_npg(), scale_fill_npg()
- AAAS, NEJM, Lancet, JAMA ...

ggsci 举例

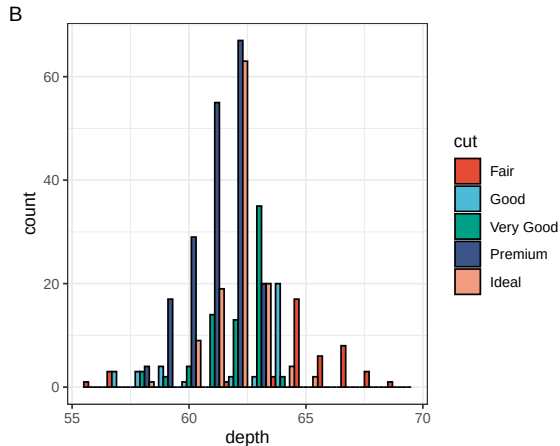
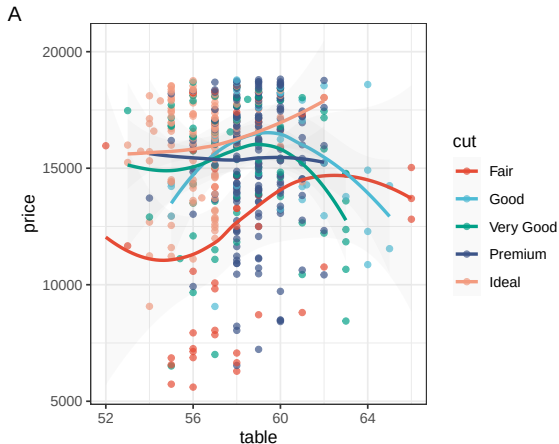
```
library("ggsci")
library("ggplot2")
library("gridExtra")
data("diamonds")
p1 <- ggplot(
  subset(diamonds, carat >= 2.2),
  aes(x = table, y = price, colour = cut)
) +
  geom_point(alpha = 0.7) +
  geom_smooth(method = "loess", alpha = 0.05, size = 1, span = 1) +
  theme_bw() + labs( tag = "A" )

p2 <- ggplot(
  subset(diamonds, carat > 2.2 & depth > 55 & depth < 70),
  aes(x = depth, fill = cut)
) +
  geom_histogram(colour = "black", binwidth = 1, position = "dodge") +
  theme_bw() + labs( tag = "B" )
```

要点: 使用了 library(gridExtra)

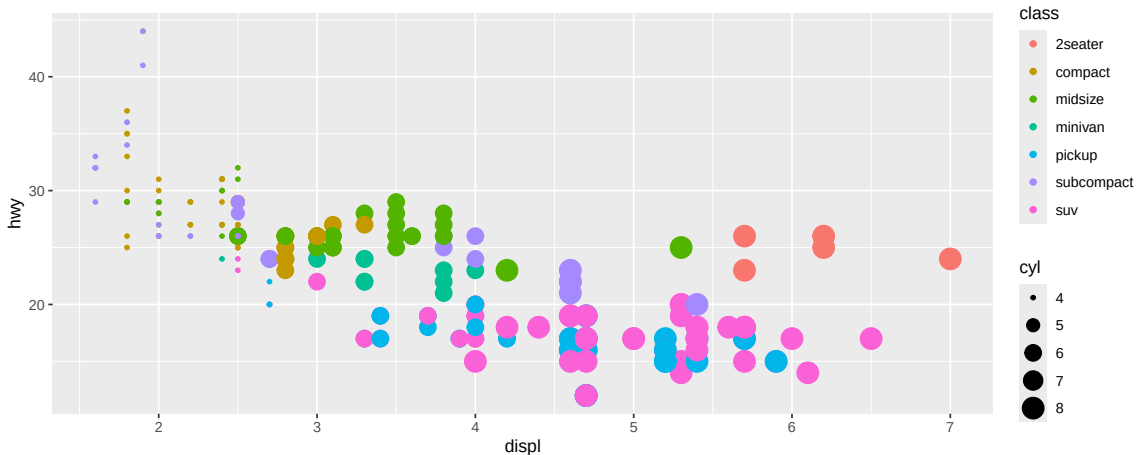
ggsci 结果, Nature Style !!

```
p1_npg <- p1 + scale_color_npg()
p2_npg <- p2 + scale_fill_npg()
grid.arrange(p1_npg, p2_npg, ncol = 2)
```



2.3 size

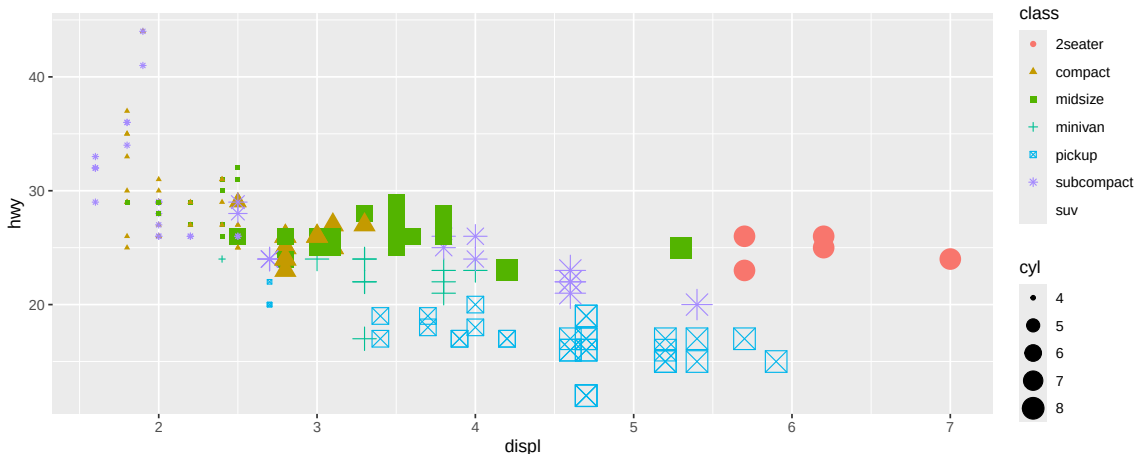
```
mpg %>% ggplot( aes(displ, hwy) ) +  
  geom_point( aes( size = cyl, colour = class ) );
```



● size : 用 cyl 列的值确定大小

2.4 shape

```
mpg %>% ggplot( aes( displ, hwy) ) +  
  geom_point( aes( size = cyl, colour = class, shape = class ) );
```



问题: size, colour 等参数可在 aes() 内部, 也可在外部, 有什么区别???

在内部时, 以指定列的值确定大小, 或按 factor 的数量确定颜色、形状的数量:

```
mpg %>% ggplot( aes(displ, hwy) ) +  
  geom_point( aes( size = cyl, colour = class, shape = class ) );
```

在外部时, 则以指定值为准:

```
mpg %>% ggplot( aes(displ, hwy) ) +  
  geom_point( size = 2, colour = "green", shape = 21 );
```

注: size 等接受的值, 长度要么为 1, 要么与行数等长;

ggplot2 要素 3: 坐标系统

线性坐标系统

- `coord_cartesian()`,
- `coord_flip()`,
- `coord_fixed()`

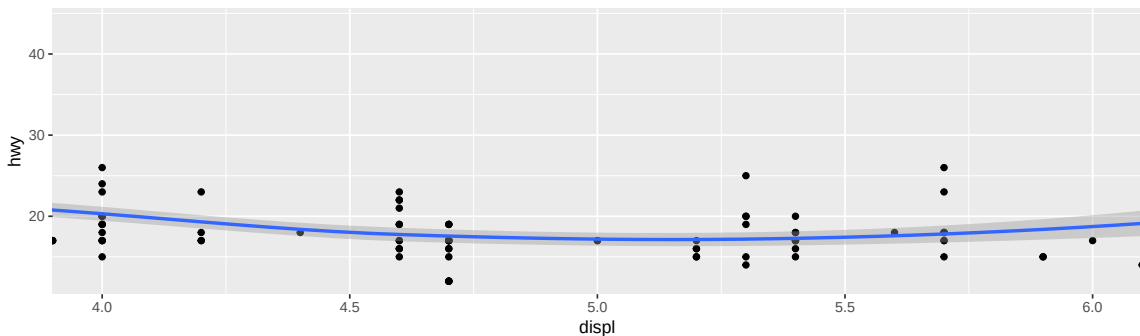
非线性坐标系统

- `coord_trans()`
- `coord_polar()`
- `coord_map()`

3.1 线性坐标系, coord_cartesian()

默认的坐标系, 可使用 `xlim`, `ylim` 等参数, 实现缩放局部

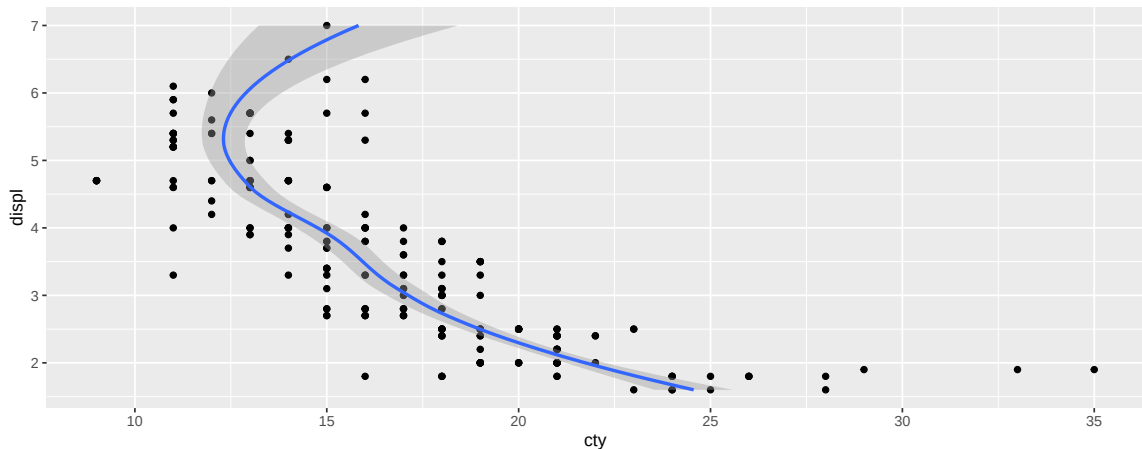
```
p1 <- ggplot(mpg, aes(displ, hwy)) +  
  geom_point() + geom_smooth()  
  
p1 + coord_cartesian(xlim = c(4, 6))
```



3.1 线性坐标系统, coord_flip()

exchange the x and y axes

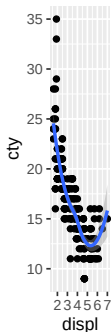
```
ggplot(mpg, aes(displ, cty)) +  
  geom_point() + geom_smooth() +  
  coord_flip()
```



3.1 线性坐标系统, coord_fixed()

用特定的数值比例 (ratio, 即: y 和 x 的真实值) 作图, 如下图所示:

```
ggplot(mpg, aes(displ, cty)) +  
  geom_point() + geom_smooth() +  
  coord_fixed(ratio = 1) ## ratio =  $y/x$ 
```

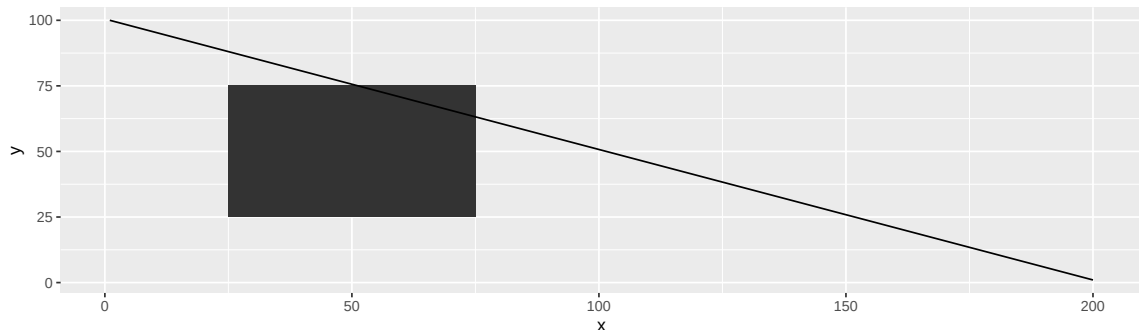


3.2 非线性, coord_trans()

x, y 轴的转换: 先生成原图

```
rect <- data.frame(x = 50, y = 50)
line <- data.frame(x = c(1, 200), y = c(100, 1))
base <- ggplot(mapping = aes(x, y)) +
  geom_tile(data = rect, aes(width = 50, height = 50)) +
  geom_line(data = line)

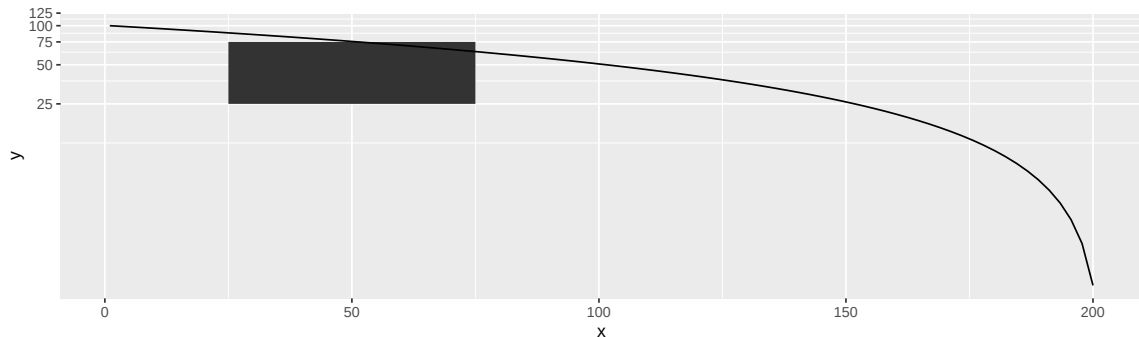
base ; ## 画出来
```



3.2 非线性, coord_trans(), cont.

$\log_{10}(y)$:

```
base + coord_trans(y = "log10");
```

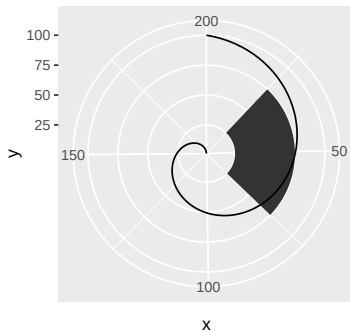


`coord_trans()` 的其它参数:

- `limx`, `limy`: 限制 `xy` 的显示范围

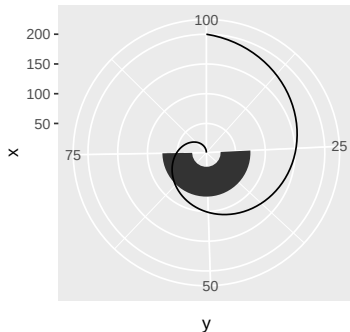
3.2 非线性, `coord_polar()`

```
base + coord_polar(); ## 默认为 coord_polar("x")
```



3.2 非线性, coord_polar(), cont.

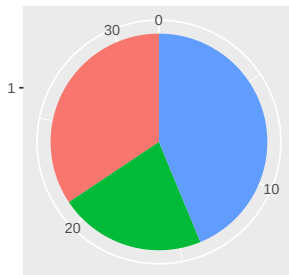
```
base + coord_polar("y");
```



3.2 非线性, coord_polar(), 柱图变饼图

```
base <- ggplot(mtcars, aes(factor(1), fill = factor(cyl))) +
  geom_bar(width = 1) + theme(legend.position = "none") +
  scale_x_discrete(NULL, expand = c(0, 0)) +
  scale_y_continuous(NULL, expand = c(0, 0))

base + coord_polar(theta = "y") ## 变饼图
```

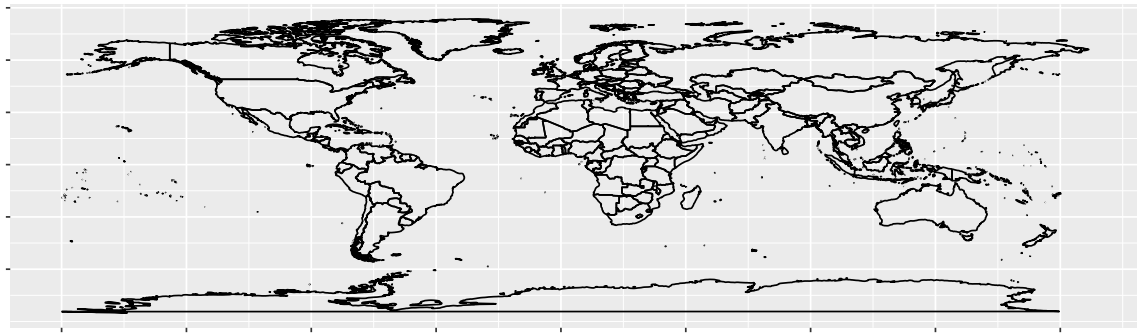


3.2 非线性, coord_map()

世界地图

```
world <- map_data("world")
worldmap <- ggplot(world, aes(long, lat, group = group)) +
  geom_path() +
  scale_y_continuous(NULL, breaks = (-2:3) * 30, labels = NULL) +
  scale_x_continuous(NULL, breaks = (-4:4) * 45, labels = NULL)

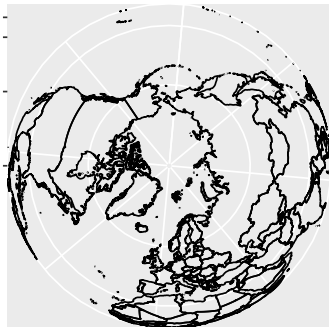
worldmap; ## 画出来
```



3.2 非线性, coord_map(), cont.

球形图

```
library(mapproj);  
worldmap + coord_map("ortho");
```



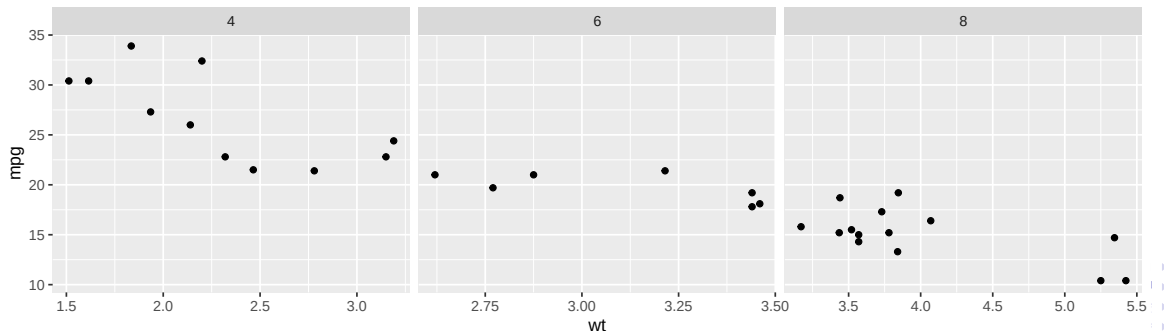
更多内容可以见《ggplot2: elegant graphics for data analysis》一书的第 7 章。

ggplot2 要素 4: faceting ...

Faceting generates small multiples each showing a different subset of the data.

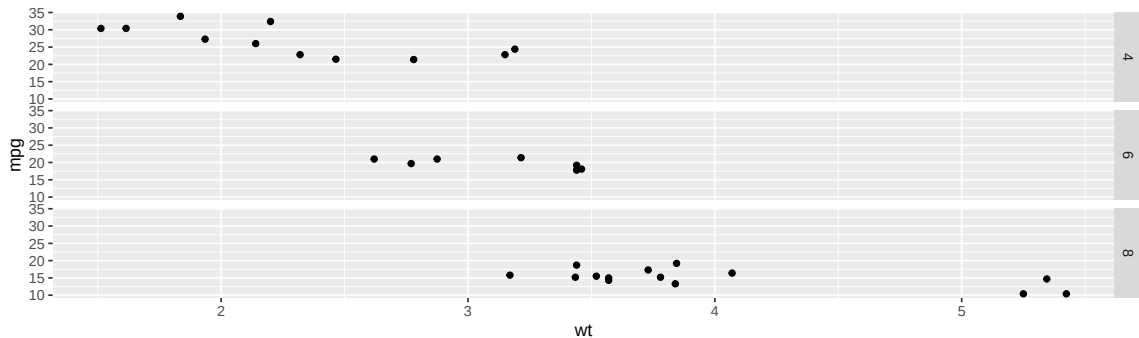
`facet_grid( <by_row> ~ <by_col> )` 汽缸、车重与燃油效率间的关系

```
ggplot( mtcars, aes( x = wt, y = mpg ) ) +  
  geom_point() +  
  facet_grid( . ~ cyl, scales = "free" );
```



faceting , cont.

```
ggplot( mtcars, aes( x = wt, y = mpg ) ) +
  geom_point() +
  facet_grid( cyl ~ . );
```

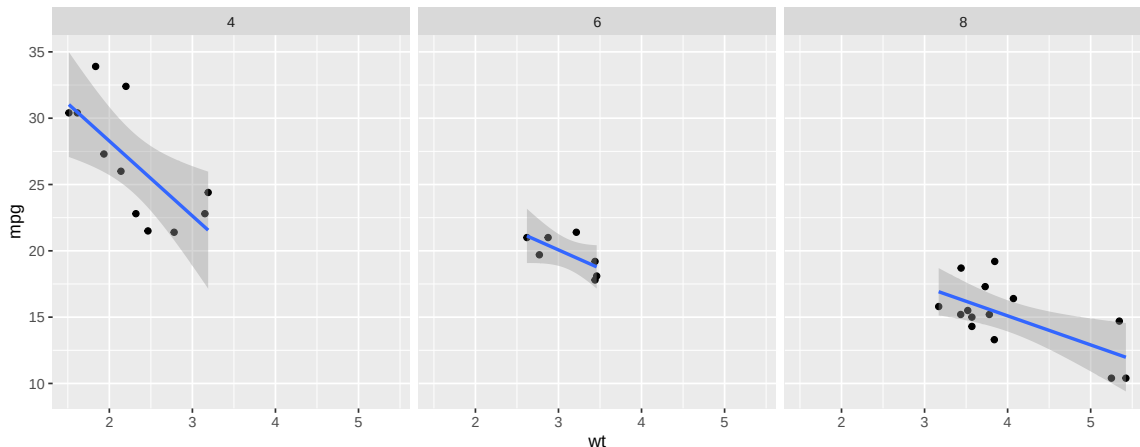


注意:

作图相关概念: panel, strip, axis, tick, tick label, axis label

facet_grid , cont.

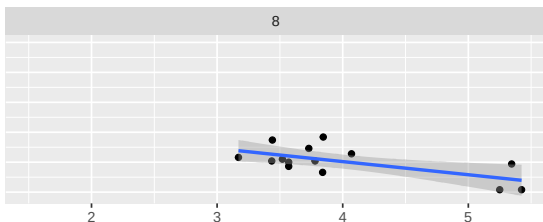
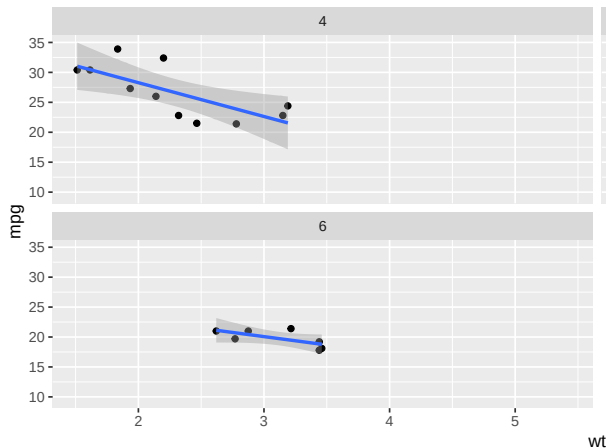
```
ggplot( mtcars, aes( x = wt, y = mpg ) ) +
  geom_point() + geom_smooth( method = "lm" ) +
  facet_grid( . ~ cyl );
```



facet_wrap

指定行、列数和方向

```
ggplot( mtcars, aes( x = wt, y = mpg ) ) +
  geom_point() + geom_smooth( method = "lm" ) +
  facet_wrap( . ~ cyl , ncol = 2, dir = "v" );
```



parameters of facet_wrap

```
facet_wrap(  
  facets,  
  nrow = NULL,  
  ncol = NULL,  
  scales = "fixed",  
  shrink = TRUE,  
  labeller = "label_value",  
  as.table = TRUE,  
  switch = NULL,  
  drop = TRUE,  
  dir = "h",  
  strip.position = "top"  
)
```

ggplot2 小结

layered grammar (图层语法) 的成分

- 图层 (`geom_XXX`)
- scale (`scale_XXX`)
- 坐标系统
- faceting (`facet_XXX`)

what's next

- no homework, yet
- will be continued at **part 2**